Closing the Style Gap: Measuring, Explaining, and Guiding the Post-Editing of LLM-Generated Text Toward Personal Authorial Voice

Dange Pooja Sanjay - B24DS006, Samarth Goyal - B24DS026, Harshita Agarwal - B24DS501

I. Project Details

Criterion followed: Data/results are available (via the authors' public repository), and our team addresses the limitations identified in the source paper by proposing and building novel analysis and tooling on top of it. Specifically, we follow the criterion *"If code/data available, address limitation and proposed solution"*, since [1] release their full set of control, LLM-generated, and post-edited texts from all 81 participants at https://github.com/ctbaumler/personal_style_postedit, but release no modeling code, guidance tools, or process-level (edit-trajectory) data. Our project both (a) runs new quantitative analyses on the released text data and (b) builds new tooling that the original study did not attempt.

Team Composition and Individual Contributions:

- Member 1: Dange Pooja Sanjay, B24DS006, dangeps@iitbhilai.ac.in Contribution - Module A1, B and E
- Member 2: Samarth Goyal, B24DS026, samarthg@iitbhilai.ac.in, Contribution - Module A3, C and D.
- Member 3:Harshita Agrawal, B24DS501, harshitaag@iitbhilai.ac.in, Contribution - Module A2 B and D.

Note: Clearly specify which part of the project each member is responsible for. (Individual ownership of Modules A–G below is suggested as one natural split of work.)

II. Problem Statement

A. Context and Motivation

[1] study whether people can post-edit LLM-generated drafts (wedding vows, apology letters, eulogies, etc.) to reclaim their personal writing style. Using LUAR-based stylistic-similarity embeddings [2], they show that post-editing significantly increases similarity to a participant's own unassisted writing (H1a) and decreases similarity to LLM-generated text (H1b). However, three findings expose an unsolved problem:

- **The gap never closes.** Post-edited text remains *significantly more* stylistically similar to LLM-generated text than to the participant's own control text (H1c, Hedges' $g = -1.43$), and is less stylistically diverse than fully human writing (H2a).
- **Users cannot see the gap.** Perceived stylistic self-similarity and model-measured self-similarity are only weakly correlated (H3, $r = 0.244$), and participants rated post-edited text as *equally* representative of their style as their own control text ($p = .9062$) even though the embeddings disagreed.
- **Editing is unguided.** Participants were given a blank draft and told to "make it sound like you," with no feedback on which spans still carried LLM stylistic fingerprints, no personalization of the initial draft to their own voice, and no record of *how* they got from draft to final text – only the before/after snapshots were analyzed.

The authors themselves flag this as an open direction: "future work could explore interventions that surface residual LLM stylistic features, allowing for more precise post-editing" (§7 of [1]).

B. Why This Matters in the ML/NLP Domain

As LLM-assisted writing becomes ubiquitous in personal and professional communication [3], stylometric and authorship-representation research [2], [4] needs to move from purely *measuring* style drift to *actively closing* it. This sits at the intersection of authorship representation, human-AI co-writing [5], AI-generated text detection [6]–[8], and interactive/explainable NLP tooling – a combination that is under-explored relative to each sub-area individually.

C. Scope of Our Work

We do *not* attempt to re-run the full 81-participant human study. Instead, we treat the paper's released dataset as ground truth and pursue two complementary tracks:

1) **Diagnostic analyses** that explain *why* and *under what conditions* the style gap in H1c/H2a is larger or smaller (length, vocabulary richness, cross-validation robustness of the similarity metric itself).
2) **Tooling** that intervenes on the editing pipeline itself – style-aware draft generation, real-time AI-likeness feedback, personalized edit recommendations, and edit-process logging – to test whether the residual-LLM-style gap and the perception–measurement gap can be narrowed.

A voice/prosody-based style-capture layer is included as an optional stretch module (§III, Module F), scoped separately since it requires an additional audio data source.

III. Methodology

We organize our work into seven modules (A–G). Modules A–C form the core deliverable; Modules D–G are extensions

that build progressively on the core style-similarity pipeline. Figure 1 sketches the overall pipeline: released paper data feeds diagnostic analyses (A) and a labeled AI-vs-human corpus (C), which in turn power the interactive tools (B, C) and the style-conditioned generator (D).

A. Module A: Diagnostics on Stylistic Similarity

Reproducing the paper's LUAR-based similarity pipeline [2] as our measurement backbone, we run three new analyses that the original paper does not report:

- **A1 – Similarity vs. length.** For every treatment document, we compute LUAR self-similarity (to the participant's control text) and LLM-similarity as a function of (a) raw word/token count and (b) number of words changed during post-editing (edit distance / WER, as already computed by the authors). We fit regression and repeated-measures correlation models (mirroring the authors' own statistical framework, §A.2 of [1]) to test whether longer edits or longer documents systematically inflate or deflate similarity scores – a possible confound the original analysis does not control for.
- **A2 – Similarity vs. vocabulary richness.** We compute lexical-diversity metrics (type-token ratio, MTLD, Yule's K) for each participant's control writing and correlate these with (i) how much their post-edited text improves in self-similarity and (ii) how much editing effort (WER) was required. Hypothesis: writers with richer/more idiosyncratic vocabularies may need less editing to sound authentic than writers with more generic vocabularies – since LLM output is comparatively lexically generic [2], [6].
- **A3 – Held-out validation of the similarity metric (Stretch Entension)** The original paper computes similarity scores using the full dataset without a train/test split for the metric itself. We partition the 81 participants (e.g., 70/30, participant-disjoint to avoid leakage) into a fit set and a held-out set: any component we calibrate (e.g., a re-weighted or fine-tuned similarity scorer, Module A2's regression) is fit only on the fit set and evaluated for generalization on the held-out set. This directly tests the robustness/reproducibility of the paper's conclusions (H1–H2) rather than taking the reported effect sizes at face value.

B. Module B: Personalized Edit Recommendation Engine

Problem addressed: the paper gives participants no guidance on what to edit (§6.4 shows highly variable, unguided editing strategies). **Approach:** using the released before/after text pairs, we align each participant's LLM draft and final post-edited text with a sequence-diff algorithm (e.g., difflib / Myers diff) to extract an edit set (substitutions, deletions, insertions) per participant. We cluster recurring edit *types* (e.g., "removes fancy/low-frequency words such as 'delve', 'explore' ", "adds contractions", "splits long sentences", "removes em dashes") using rule-based lexical features plus a lightweight classifier (e.g., logistic regression / small transformer over edit spans). At inference time, given a *new* LLM draft and a user's historical edit profile, the system ranks candidate edit locations by predicted relevance to that user, functioning as a recommender system over edit operations rather than over items.

C. Module C: Real-Time Visual AI-Likeness Feedback

Problem addressed: §6.3 of the paper shows participants cannot tell which parts of their post-edited text still read as AI-generated (H3's weak correlation). **Approach:** we fine-tune a span-level AI-vs-human classifier (e.g., DeBERTa-v3 or RoBERTa-base) on the paper's own labeled pairs – LLM-generated spans vs. human control spans are a naturally-supervised signal for this exact task – augmented with the RAID [7] and HC3 [8] corpora for broader robustness. The model outputs a per-token/per-sentence AI-likelihood score, rendered in a browser-based editor as a red/yellow/green heatmap that updates as the user types, giving continuous feedback instead of a single end-of-task label.

D. Module D: Cross-Topic, Attention-Based Style Capture

Using an attention-based style encoder (in the spirit of [9], applied over LUAR/CISR-style sentence embeddings [2], [4]), we extract a style vector from a user's writing on one topic (e.g., their control text for "catch-up letter") and use it to condition draft generation for a *different* topic (e.g., "apology letter") via style-conditioned prompting or a lightweight retrieval-augmented generation step. We then measure whether the *initial, unedited* draft is already closer to the user's style than the paper's generic zero-shot GPT-o4-mini draft, which would reduce the post-editing burden captured in Module B/D.

E. Module E: Document-Level AI-Content Quantification (Extension)

A coarser, document-level companion to Module C: given a finished document, output an estimated % AI-originated vs. % human-edited content, using the same classifier backbone as Module C aggregated over spans, calibrated against the known ground truth in the released dataset (where the LLM draft and the human edits are both known).

F. Module F: Voice/Prosody-Informed Style Capture (Stretch Extension)

Problem addressed: Reza et al. (cited in [1], §2.2) find writers most resist AI help specifically at the "translating ideas into language" stage – i.e., converting a plan into wording. We explore whether letting users *speak* their planned details (as in the paper's "planning" phase, Figure 2 of [1]) and extracting prosodic/emotional features (pitch, pace, pausing, sentiment/emotion class) can inform the tone of the generated draft, rather than using only typed bullet points as in the original study. This is scoped as a separate, optional module since it requires its own audio dataset and consent process (§IV).

G. Module G: Edit-Process Logging and Analysis (Stretch Extension)

Problem addressed: the paper analyzes only the final before/after text pair (§B.7.3 notes this is a known limitation: dense vs. diffuse edit *patterns* were only summarized statically). **Approach:** we build a lightweight browser text-editor instrumented to log every keystroke, selection, deletion, and pause with timestamps, reconstructing the full edit trajectory (not just the final diff). Using pilot data collected from classmates/volunteers performing the same post-editing task the paper used, we analyze: (i) order of edits (are AI-flagged spans from Module C fixed first?), (ii) time-to-first-edit and edit velocity, and (iii) whether edit *order/process* predicts final self-similarity better than edit *count* alone (extending §B.7.3's number-vs-length analysis into the time dimension).

H. Summary Pipeline

1) Ingest released dataset (control, LLM-draft, post-edited text per participant/task).
2) Module A: statistical diagnostics on similarity vs. length/vocabulary + held-out validation.
3) Fine-tune the AI-content classifier (Module C) on labeled pairs, which also underlies Module E's document-level scoring.
4) Modules B, C (and optionally G): interactive editing tool (recommendations + live feedback + process logging), piloted on volunteer writers performing the paper's own tasks.
5) Module D (and optionally F) : style-conditioned generation to reduce the editing burden at the source.
6) Evaluate all modules against the paper's original H1/H2/H3 metrics to test whether the gaps they report shrink.

IV. Resource and Dataset Availability

A. Primary Dataset (Available)

- **Name/source:** "personal_style_postedit" dataset released by [1] at https://github.com/ctbaumler/personal_style_postedit.
- **Size/structure:** Text from 81 participants across 8 writing-task categories (Table 9 of the paper): control (unassisted) text, original LLM-generated (GPT-o4-mini) drafts, and post-edited treatment text, plus task-level and post-survey perception ratings – roughly 300+ control documents and 250+ treatment documents in total across tasks.
- **Preprocessing:** sentence/token segmentation, alignment of LLM-draft/post-edit pairs via diff (Module B), computation of LUAR embeddings [2] and lexical-diversity statistics (Module A), participant-disjoint train/held-out split (Module A3).
- **Justification:** this is the exact dataset the paper's own hypotheses (H1–H3) were computed on, making it the only dataset that lets us directly test whether our interventions reduce the gaps the authors report, and it is the natural source of supervised AI-vs-human labels for Modules C/E (LLM draft = AI-labeled span, control text = human-labeled span).

B. Secondary/Augmentation Datasets (Available, for robustness)

- **RAID** [7] and **HC3** [8] – large, multi-model AI-vs-human text corpora, used to make the Modules C/E classifier robust beyond the (relatively small, single-domain) supervision available in the primary dataset.

C. Dataset for Module F (Optional/Stretch, To Be Collected or Public)

- **Public options:** Mozilla Common Voice [10] (general prosody), IEMOCAP [11] or MELD [12] (emotion-labeled speech) for pretraining/validating the emotion-extraction component.
- **Task-specific data:** since none of the public corpora contain "planning a personal writing task aloud," task-matched data would need to be self-collected via consented short voice memos from volunteers, following the same anonymization/consent principles as §9 of [1].

D. Dataset for Module G (Optional/Stretch, To Be Collected)

No edit-*process* (keystroke-level) data exists publicly for this task. We will build a small instrumented web editor and collect pilot edit-trajectory data from a small volunteer pool (e.g., classmates, $n \approx 15$–20) performing the same category of writing tasks used in the paper (Table 9), under the same "do not include real personal details" guidance the original authors used (§9, Ethical Considerations of [1]). This is explicitly a small-scale pilot, not a claim of statistical power comparable to the original $n = 81$ study.

E. Hardware

- Fine-tuning the span-level AI-detection classifier (Modules C/E) and the style encoder (Module D): a single mid-range GPU (e.g., Google Colab/Kaggle T4/P100, or department lab GPU) is sufficient given model sizes (RoBERTa-base/DeBERTa-v3-base scale) and dataset size.
- Diagnostic analyses (Module A) and the recommendation engine (Module B) are CPU-feasible.
- **Available hardware:** the team has direct access to a laptop with a 12 GB GPU and 24 GB RAM, which meets the above requirement for fine-tuning base-scale (RoBERTa-base/DeBERTa-v3-base) models using mixed-precision training, and is sufficient as the primary compute resource for this project without relying on external GPU access.

F. Software

Python; HuggingFace `transformers` and `sentence-transformers` (for reproducing LUAR-style embeddings); `scikit-learn` / `statsmodels` (regression, repeated-measures correlation, permutation tests, mirroring §A.2 of [1]); `lexicalrichness` or `textstat`

(vocabulary-diversity metrics); a browser front end (React/JS) for the live-feedback and edit-logging editor (Modules C/G); `difflib`/Myers-diff libraries for edit alignment (Module B); optionally `openai-whisper` and `librosa` for voice/prosody extraction (Module G).

G. Fallback Plan

If GPU access or the Module F/G volunteer pilots are not feasible within the term, we will prioritize Modules A, B, and E (all computable directly from the publicly released dataset with modest compute), and scope Modules C, D, F, G as stretch goals reported qualitatively/partially.

V. Limitations Found

1) **No confound analysis on the similarity metric itself.** The paper reports overall effect sizes for H1/H2 but does not check whether document length or vocabulary richness confound the LUAR similarity score. *Addressed by Module A1/A2.*
2) **No held-out validation of the similarity pipeline.** All reported statistics are computed over the full 81-participant dataset without a train/test split for any calibrated component, limiting confidence that any refinement would generalize. *Addressed by Module A3.*
3) **Editing is unguided.** Participants received no signal about which edits actually move the needle on style, leading to highly inconsistent editing strategies (§6.4). *Addressed by Module B.*
4) **Perception–measurement gap is diagnosed but not remediated.** The paper documents (H3) that users cannot tell their text still reads as AI-generated, but proposes no tool to close this gap, explicitly leaving it to future work. *Addressed by Module C.*
5) **Only the final text is analyzed, not the editing process.** §B.7.3 acknowledges edit density/length were only summarized statically post-hoc; no temporal/keystroke data was collected. *Addressed by Module G.*
6) **Drafts are not personalized to the user's own style at generation time** – §B.1 confirms the LLM draft does not pick up on the participant's stylistic cues, meaning every participant starts editing from the same generic style baseline. *Addressed by Module D.*
7) **No user-facing, aggregate AI-content signal.** Detection scores (Pangram) were used only for research analysis, not surfaced to participants. *Addressed by Module E.*
8) **Only text-based planning is considered.** Prior work cited by the authors shows resistance to AI help specifically at the idea-to-language translation stage; the paper does not test whether a different input modality (voice) changes this. *Addressed by Module F, as a stretch goal.*

What differentiates our work: rather than re-running the human study, we (a) stress-test the paper's own statistical pipeline for confounds and generalization (Module A), and (b) build the first tooling layer that acts *during* editing rather than only measuring *after* editing (Modules B–D), directly operationalizing the paper's own stated future-work direction (§7: "surface residual LLM stylistic features"). Modules D–F extend the study's scope (personalized generation, aggregate AI-scoring, voice input) beyond what the original single-shot, text-only, zero-shot pipeline attempted.

Improvements targeted: (i) a demonstrable reduction in the H1c gap (post-edited-to-LLM similarity vs. post-edited-to-self similarity) when drafts are style-conditioned (Module D) or edits are guided (Module B/C); (ii) a more reliable, held-out-validated similarity metric (Module A3); (iii) new empirical evidence on *which* editing behaviors (not just how many edits) predict successful style recovery (Module G).

VI. Expected Outcomes

A. Analytical Deliverables (Module A)

- Correlation/regression results quantifying how self-similarity and LLM-similarity scores vary with document length and vocabulary richness (with effect sizes and confidence intervals, mirroring the paper's own reporting style, §A.2 of [1]).
- A held-out-validated version of the similarity analysis, reporting whether the original H1/H2 effect directions and significance replicate on unseen participants.

B. System Deliverables (Modules B, C, E)

- A working prototype combining: (i) real-time red/yellow/green AI-likeness highlighting, (ii) ranked, personalized edit suggestions, and (iii) a document-level %-AI-content score.
- Target metric: AI-detection classifier (Modules C/E) achieving competitive span-level F1 against RAID/HC3 baselines [7], [8], and, on the paper's own held-out control/LLM-draft pairs, high separability (AUROC) consistent with reported Pangram performance [6].

C. Extension Deliverables (Modules D, F)

- A style-conditioned draft generator (Module D) with a before/after comparison showing whether *unedited* style-conditioned drafts already score closer to a user's self-similarity than the paper's generic zero-shot baseline.
- (Stretch) A voice-informed drafting prototype (Module F) with a short qualitative/pilot evaluation.

D. Final Outputs

- A written project report structured around the paper's own H1/H2/H3 framework, showing before-vs-after (paper vs. our interventions) comparisons.
- A working application (Modules B/C/E) usable on new writing tasks.
- A public code/analysis repository documenting all modules, reproducible from the released dataset.

References

[1] C. Baumler, C. Bao, H. Nghiem, X. Yang, M. Carpuat, and H. Daumé III, "Can you make it sound like you? post-editing llm-generated text for personal style," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2026, pp. 43 867–43 895.

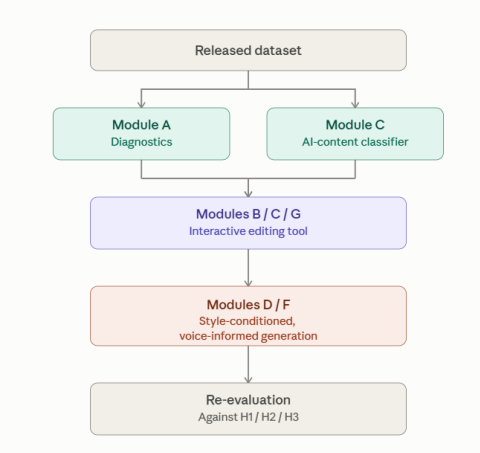


Fig. 1. Overall project pipeline. The released dataset feeds Module A's diagnostic analyses and the labeled corpus behind Module C's AI-content classifier. Both inform the interactive editing tool (Modules B, C, and G), whose output guides style-conditioned and voice-informed draft generation (Modules D and F). All interventions are then re-evaluated against the original paper's H1, H2, and H3 hypotheses.

[2] R. A. Rivera-Soto, O. E. Miano, J. Ordonez, B. Y. Chen, A. Khan, M. Bishop, and N. Andrews, "Learning universal authorship representations," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 913–919.
[3] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
[4] A. Wegmann, M. Schraagen, and D. Nguyen, "Same author or just same topic? towards content-independent style representations," in *Proceedings of the 7th Workshop on Representation Learning for NLP*, 2022, pp. 249–268.
[5] M. Reza, J. Thomas-Mitchell, P. Dushniku, N. Laundry, J. J. Williams, and A. Kuzminykh, "Co-writing with ai, on human terms: Aligning research with user demands across the writing process," *Proc. ACM Hum.-Comput. Interact.*, vol. 9, no. 7, 2025.
[6] B. Emi and M. Spero, "Technical report on the pangram ai-generated text classifier," *arXiv preprint arXiv:2402.14873*, 2024.
[7] L. Dugan, A. Hwang, F. Trhlík, A. Zhu, J. M. Ludan, H. Xu, D. Ippolito, and C. Callison-Burch, "Raid: A shared benchmark for robust evaluation of machine-generated text detectors," *arXiv preprint arXiv:2405.07940*, 2024.
[8] B. Guo, X. Zhang, Z. Wang, M. Jiang, J. Nie, Y. Ding, J. Yue, and Y. Wu, "How close is chatgpt to human experts? comparison corpus, evaluation, and detection," *arXiv preprint arXiv:2301.07597*, 2023.
[9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
[10] Mozilla Foundation, "Common voice: A massively-multilingual speech corpus," https://commonvoice.mozilla.org/, 2023.
[11] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, "Iemocap: Interactive emotional dyadic motion capture database," vol. 42, 2008, pp. 335–359.
[12] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019.

APPENDIX

This SOP was drafted with the assistance of Claude (Anthropic) for structuring, LaTeX formatting, and organizing the team's brainstormed ideas against the source paper's stated limitations; all technical proposals (Modules A–G) originated from the team's own project discussions. Source paper: [1]. Post-Drafting restructuring and reorganization done by the individual members only.

TABLE I
MAPPING OF PROPOSED MODULES TO THE PAPER'S LIMITATIONS

Module	Limitation Addressed
A	No length/vocabulary confound check; no held-out validation
B	Editing is unguided (§6.4)
C	Perception–measurement gap unremediated (H3)
D	Drafts not personalized at generation time (§B.1)
E	No user-facing AI-content signal
F	Only text-based planning considered (stretch)
G	Only final text analyzed, not editing process (§B.7.3)